

Real-Time FPGA Implementation of a QPSK Modulator and Demodulator with Simulated Channel

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Abstract

Quadrature Phase Shift Keying (QPSK) is a digital modulation technique which transmits two bits per symbol without additional bandwidth. This project implements a complete, real-time QPSK transmitter and receiver entirely on a Spartan-7 FPGA (Boolean Board, XC7S50). The system performs bit-to-symbol mapping, carrier generation, quadrature mixing, and matched low-pass filtering, and includes a simulated AWGN channel with live-adjustable noise level for testing. System performance is validated through bit-error-rate (BER) measurement and live constellation visualization streamed over UART. This report defines the problem statement, reviews relevant literature on QPSK theory and existing FPGA-based QPSK implementations, and outlines the architecture and verification plan for the mid-term and final phases. Carrier recovery via a Costas loop, symbol timing recovery via the Gardner algorithm, and a coded-versus-uncoded BER comparison are being considered as optional extensions if time permits.

1. Introduction